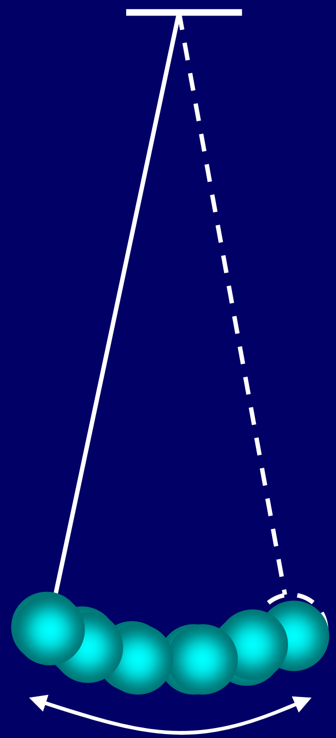
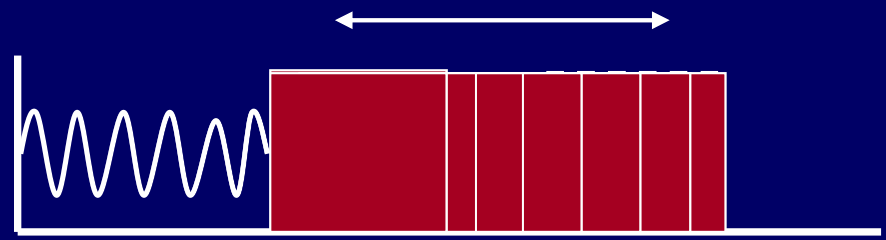
CHAP.17
Oscillation

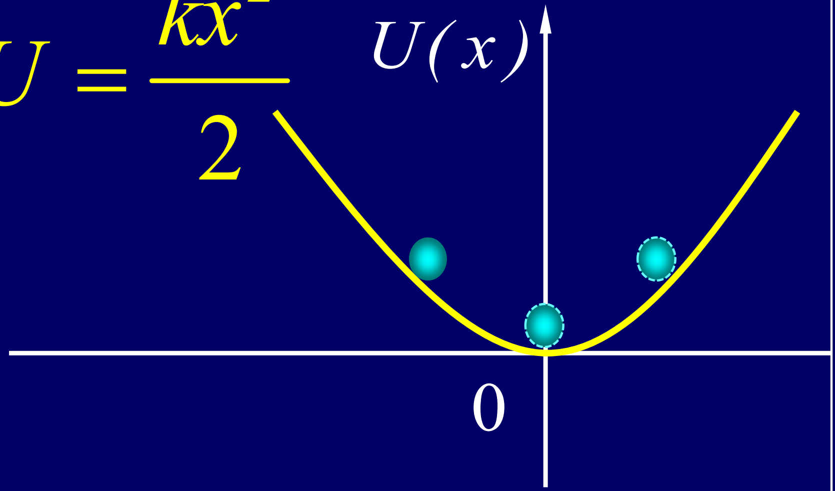
Oscillation → motion

Simple Harmonic Motion

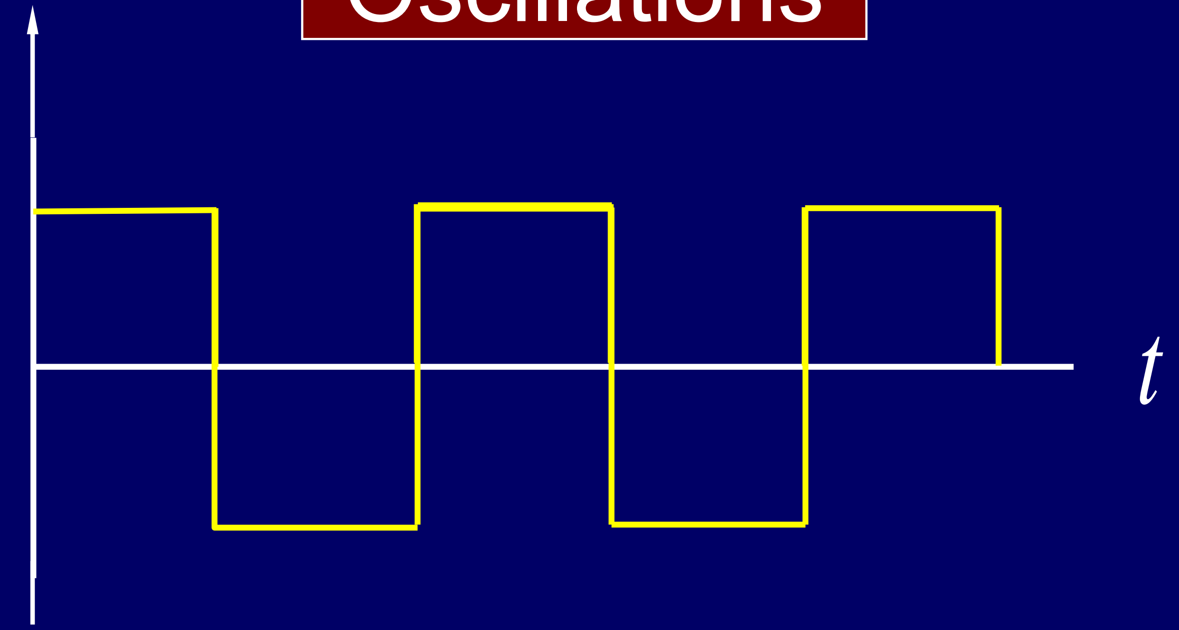


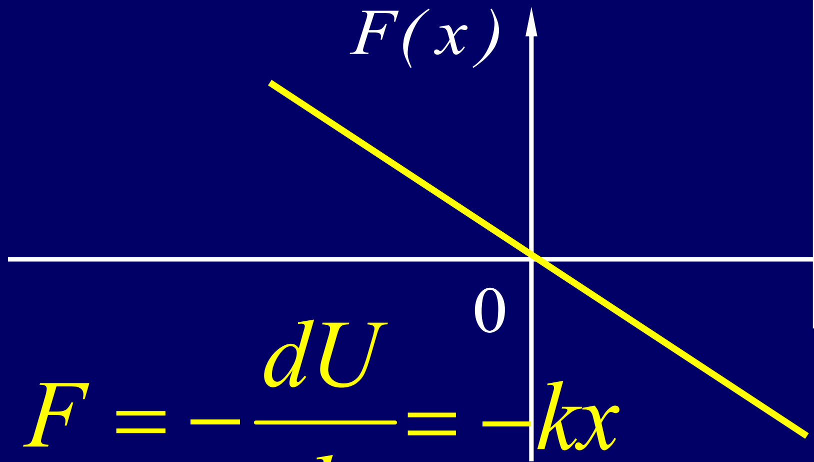


$$U = \frac{kx^2}{2}$$



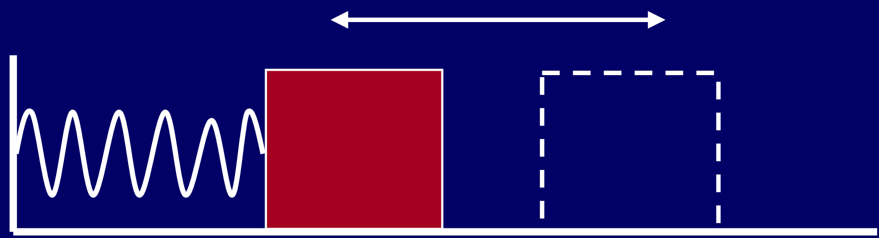
Oscillations



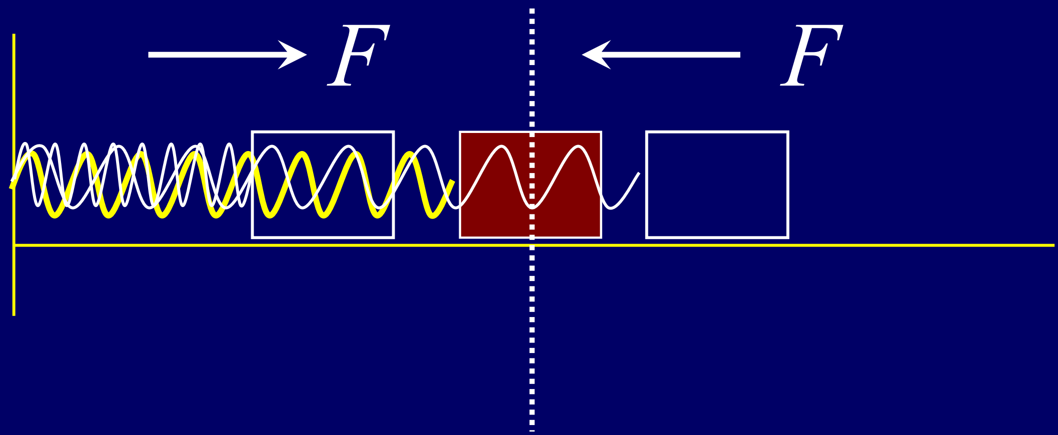


$$F = -\frac{dU}{dx} = -kx$$

Simple Harmonic Oscillation



Simple Harmonic Oscillations



$$F = -kx = ma = m\frac{d^2x}{dt^2}$$

$\boldsymbol{x}$

$$-kx = m\frac{d^2x}{dt^2}$$

$$\omega^2 = \frac{k}{m} \quad \frac{d^2x}{dt^2} + \omega^2 x = 0$$

- How to solve the equation of motion:

$$\frac{d^2x}{dt^2} = -\omega^2 x$$

$$\frac{d^2x}{dt^2} = \frac{d}{dt}(\frac{dx}{dt}) = \frac{dx}{dx}\frac{d}{dt}(\frac{dx}{dt}) = \frac{dx}{dt}\frac{d}{dx}(\frac{dx}{dt}) = \frac{1}{2}\frac{d}{dx}(\frac{dx}{dt})^2$$

$$\frac{d}{dx}(\frac{dx}{dt})^2 = -2\omega^2 x \Rightarrow (\frac{dx}{dt})^2 = -\omega^2 x^2 + \mathrm{c}^2 \Rightarrow \frac{dx}{dt} = \pm\sqrt{c^2 - \omega^2 x^2}$$

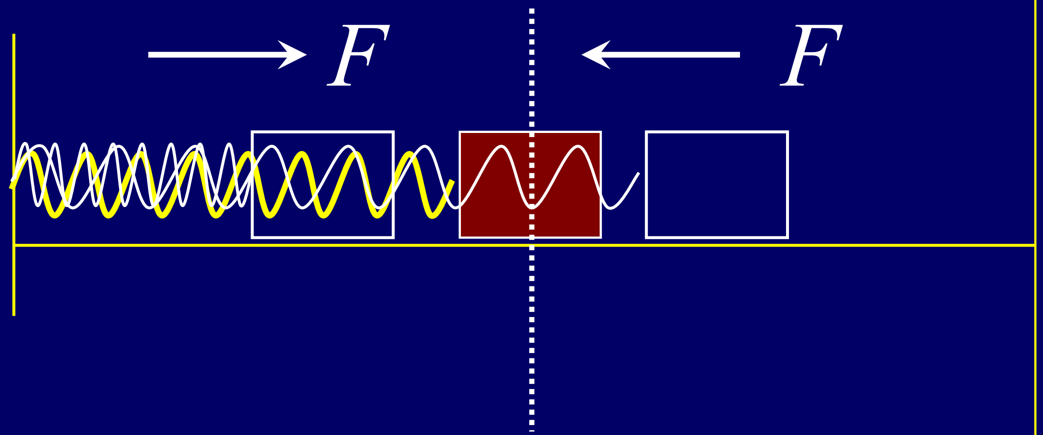
$$\frac{dx}{\sqrt{c^2 - \omega^2 x^2}} = \pm dt \Rightarrow \int \frac{d\omega x / c}{\sqrt{1 - \omega^2 x^2 / c^2}} = \mp \int \omega dt$$

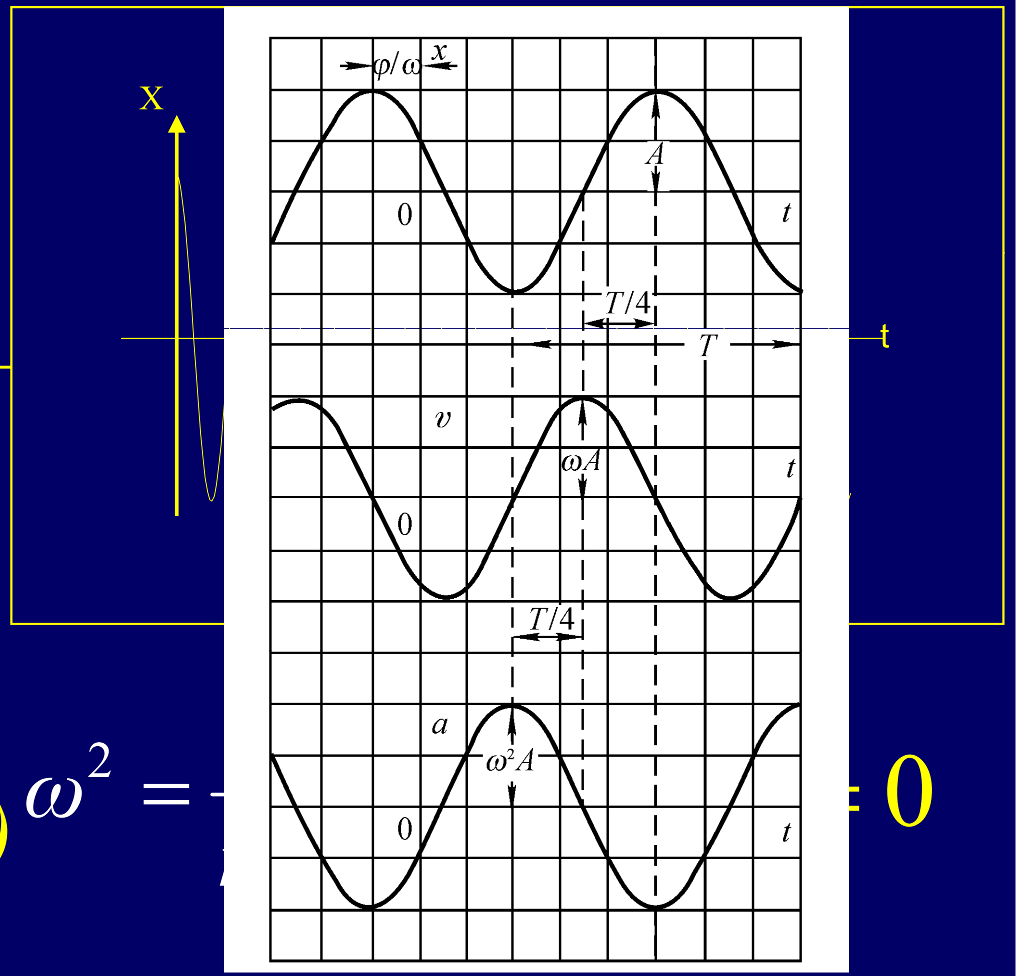
$$acr\cos(\omega x / c) = \pm \omega t + \phi$$

$$x = (c / \omega)\cos(\pm \omega t + \varphi) \Rightarrow x = x_m \, cos(\omega t + \phi)$$

Simple Harmonic Oscillations

Simple Harmonic Oscillation





$$v = \frac{dx}{dt} = -x_m \omega \sin(\omega t + \phi)$$

$$a = \frac{d^2 x}{dt^2} = -x_m \omega^2 \cos(\omega t + \phi)$$

$$x = x_m \cos(\omega t + \phi)$$

Simple Harmonic Oscillation

$$\omega = \sqrt{\frac{k}{m}}$$ **depends on the system**

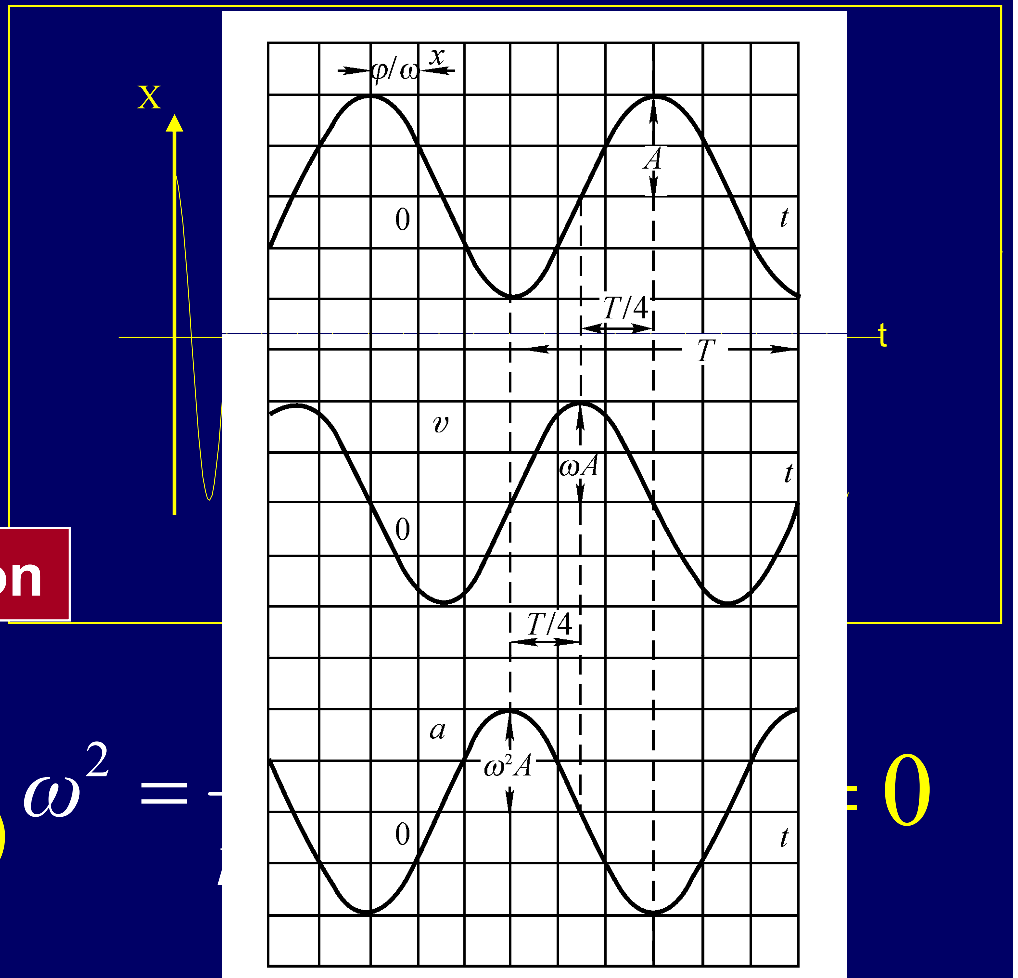
$$\omega = 2\pi f \quad "f" \ frequency$$

$$\omega = \frac{2\pi}{T} \quad "T" \text{ period}$$

x_m amplitude $\quad \omega t + \phi \quad$ *phase*

depend on the initial condition

$\phi \quad$ *phase constant*

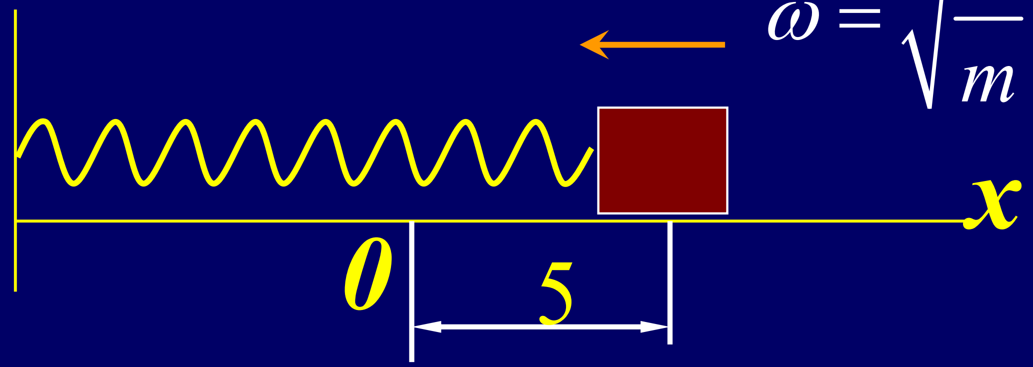


$$v = \frac{dx}{dt} = -x_m\omega \sin(\omega t + \phi)$$

$$\omega^2 = \ldots = 0$$

$$a = \frac{d^2x}{dt^2} = -x_m\omega^2 \cos(\omega t + \phi)$$

$$x = x_m \cos(\omega t + \phi)$$



$$\omega = \sqrt{\frac{k}{m}}$$

$$when \;\; t = 0, \;\; x = 5, \;\; v = 0$$

$$v = \frac{dx}{dt} = -x_m \omega \sin(\omega t + \phi)$$

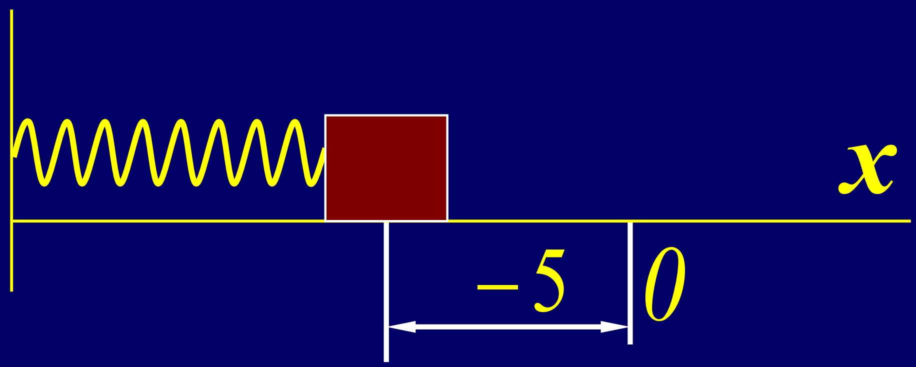
$$x = x_m \, cos(\omega t + \phi)$$

$$v\Big|_{t=0} = \frac{dx}{dt}\Big|_{t=0} = -x_m \omega \sin\phi = 0$$

$$\varphi = 0$$

$$x = x_m \cos\omega t \qquad x\Big|_{t=0} = x_m = 5$$

$$x = 5\, cos\, \omega t$$



$$when\ \ t=0,\ \ x=-5,\ \ v=0$$

$$v=\frac{dx}{dt}=-x_m\omega\sin(\omega t+\phi)$$

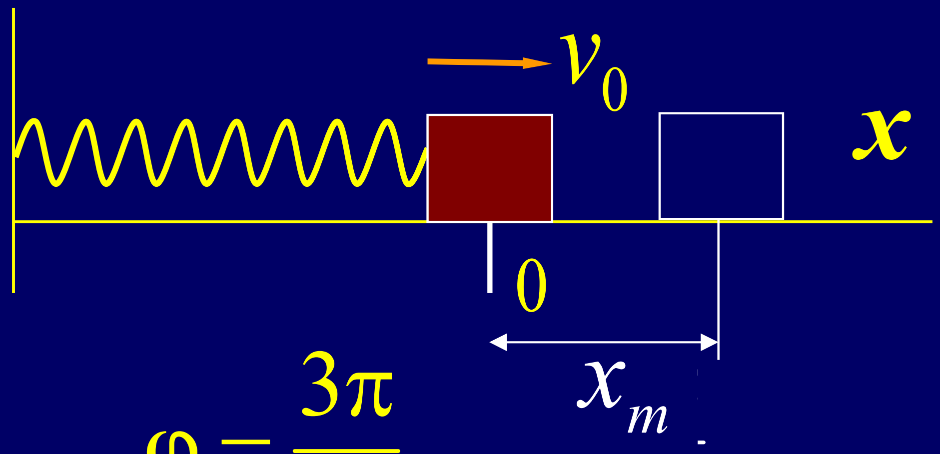
$$x=x_m\,cos(\omega t+\phi)$$

$$v\Big|_{t=0}=\frac{dx}{dt}\Big|_{t=0}=-x_m\omega\sin\phi=0$$

$$\varphi=\pi$$

$$x=x_m\cos(\omega t+\pi)\qquad x\Big|_{t=0}=-x_m=-5$$

$$x=5\cos(\omega t+\pi)$$



$$when\ \ t=0,\ \ x=0,\ \ v=v_0$$

$$x = x_m\cos(\omega t+\phi)$$

$$x\Big|_{t=0} = x_m\cos\phi = 0$$

$$\varphi = \frac{3\pi}{2}$$

$$v\Big|_{t=0} = \frac{dx}{dt}\Big|_{t=0} = -x_m\omega\sin\phi = v_0$$

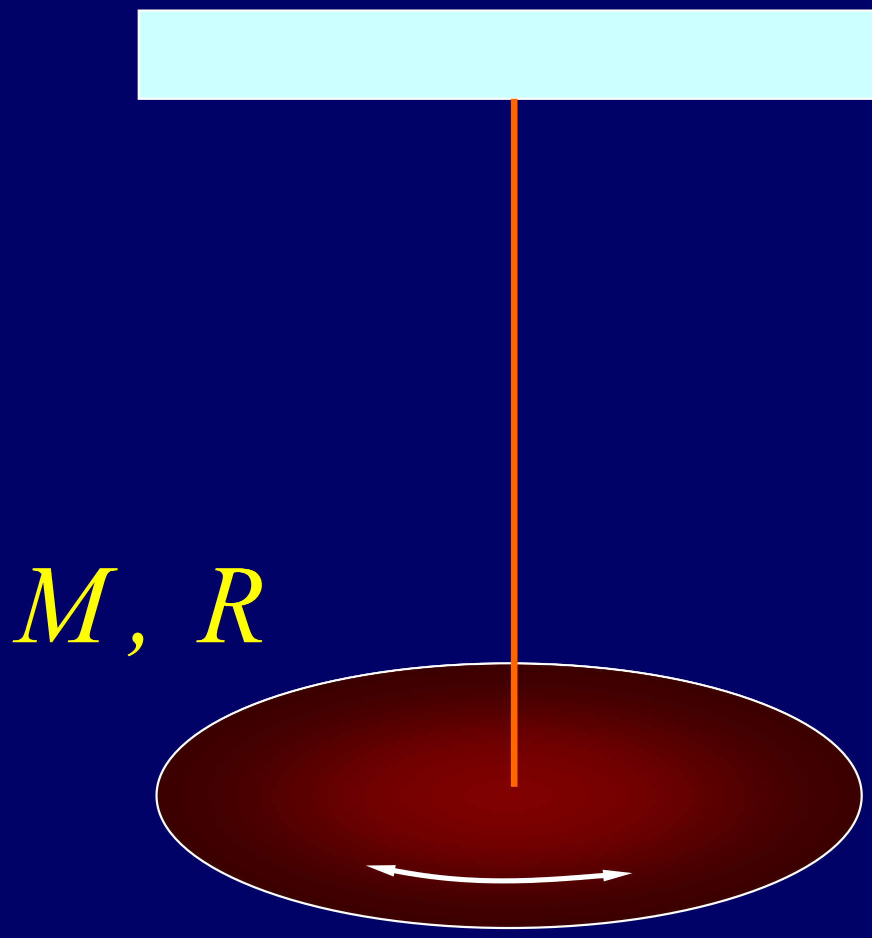
$$x_m\omega = v_0 \quad x_m = \frac{v_0}{\omega}$$

$$x = \frac{v_0}{\omega}\cos(\omega t+\frac{3\pi}{2})$$

$$\omega = \sqrt{\frac{k}{m}} \qquad x_m = v_0\sqrt{\frac{m}{k}}$$

$$U = \frac{1}{2}kx_m^2 = \frac{1}{2}mv_0^2$$

Potential Energy **Kinetic Energy**



what is ω?

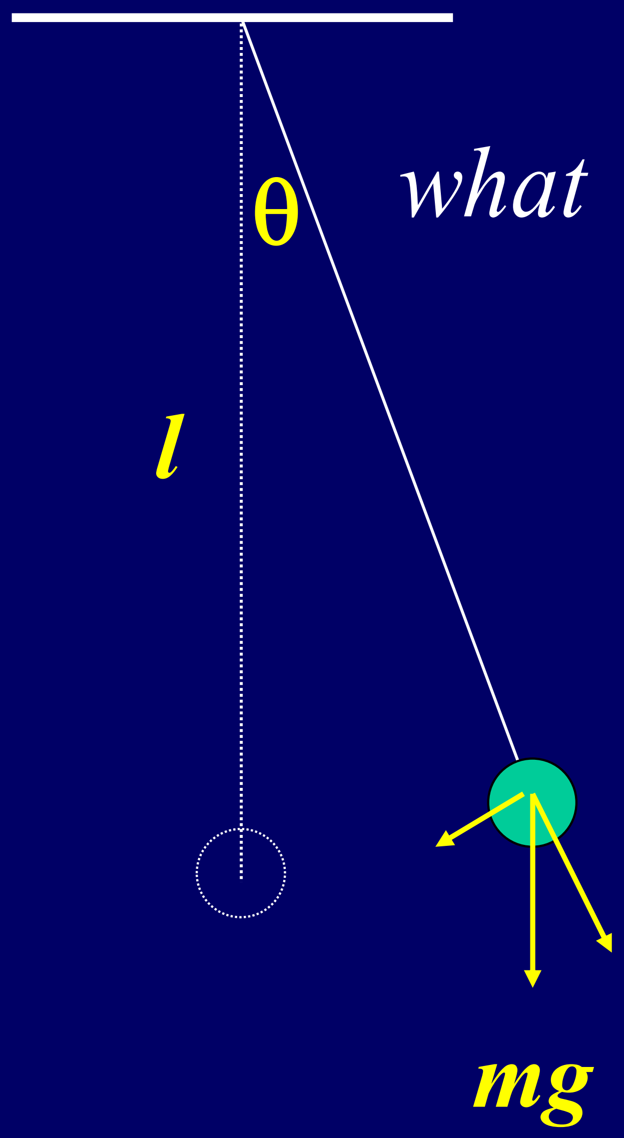
$$\tau = -k\theta$$

$$-k\theta = \left(\frac{1}{2}MR^2\right)\frac{d^2\theta}{dt^2}$$

$$\frac{d^2\theta}{dt^2} + \frac{2k}{MR^2}\theta = 0$$

$$\omega = \sqrt{\frac{2k}{MR^2}} \qquad \frac{d^2\theta}{dt^2} + \omega^2\theta = 0$$

$$\theta = \theta_m \cos(\omega t + \phi)$$



what is ω?

$$\tau = -\,mgl\sin\theta$$

$$-g\sin\theta = l\frac{d^2\theta}{dt^2}$$

$$-g\theta = l\frac{d^2\theta}{dt^2}$$

$$\frac{g}{l} = \omega^2 \qquad \frac{d^2\theta}{dt^2} + \omega^2\theta = 0$$

$$\theta = \theta_m \cos(\omega t + \varphi)$$

what is ω?

$$\tau = -mg\frac{l}{2}\sin\theta$$

$$-mg\frac{l}{2}\sin\theta = \frac{1}{3}ml^2\frac{d^2\theta}{dt^2}$$

$$-\frac{g}{2}\theta = \frac{l}{3}\frac{d^2\theta}{dt^2}$$

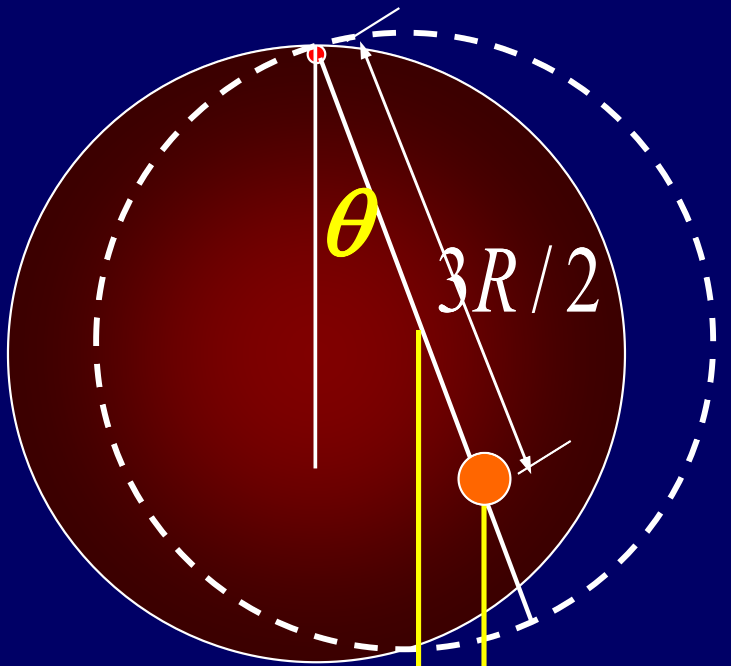
$$\frac{d^2\theta}{dt^2} + \frac{3g}{2l}\theta = 0$$

$$l' = 2l/3$$

$$\frac{g}{l'} = \frac{g}{2l/3} = \frac{3g}{2l} = \omega^2 \qquad \frac{d^2\theta}{dt^2} + \omega^2\theta = 0$$

$$\theta = \theta_m\, cos(\omega t + \phi)$$

what is ω?



$$\tau = -mgR\sin\theta$$

$$-mgR\sin\theta = \frac{3}{2}mR^2\frac{d^2\theta}{dt^2}$$

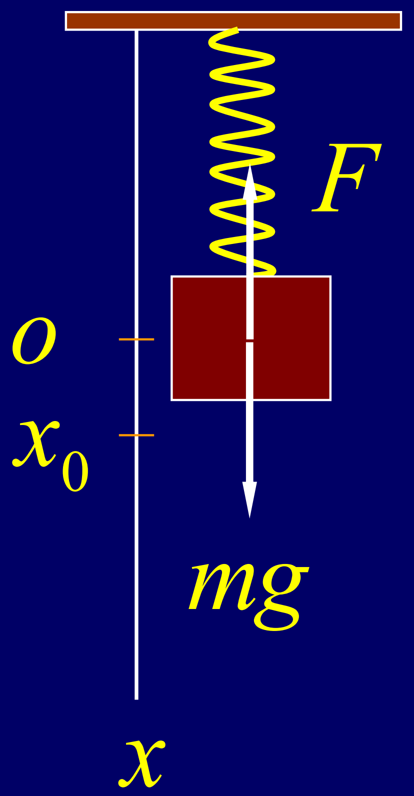
$$-g\theta = \frac{3R}{2}\frac{d^2\theta}{dt^2}$$

$$\frac{d^2\theta}{dt^2} + \frac{2g}{3R}\theta = 0$$

$$\frac{g}{L} = \omega^2 \Leftarrow \frac{g}{3R/2} = \omega^2 \Leftarrow \frac{2g}{3R} = \omega^2 \qquad \frac{d^2\theta}{dt^2} + \omega^2\theta = 0$$

mg

$$\theta = \theta_m \cos(\omega t + \phi)$$



$$mg + F = ma \qquad mg - kx = m\frac{d^2x}{dt^2}$$

$$mg - kx_0 = 0 \qquad kx_0 - kx = m\frac{d^2x}{dt^2}$$

$$mg = kx_0$$

$$-k(x - x_0) = m\frac{d^2(x - x_0)}{dt^2}$$

$$let \quad x\text{-}x_0 = x'$$

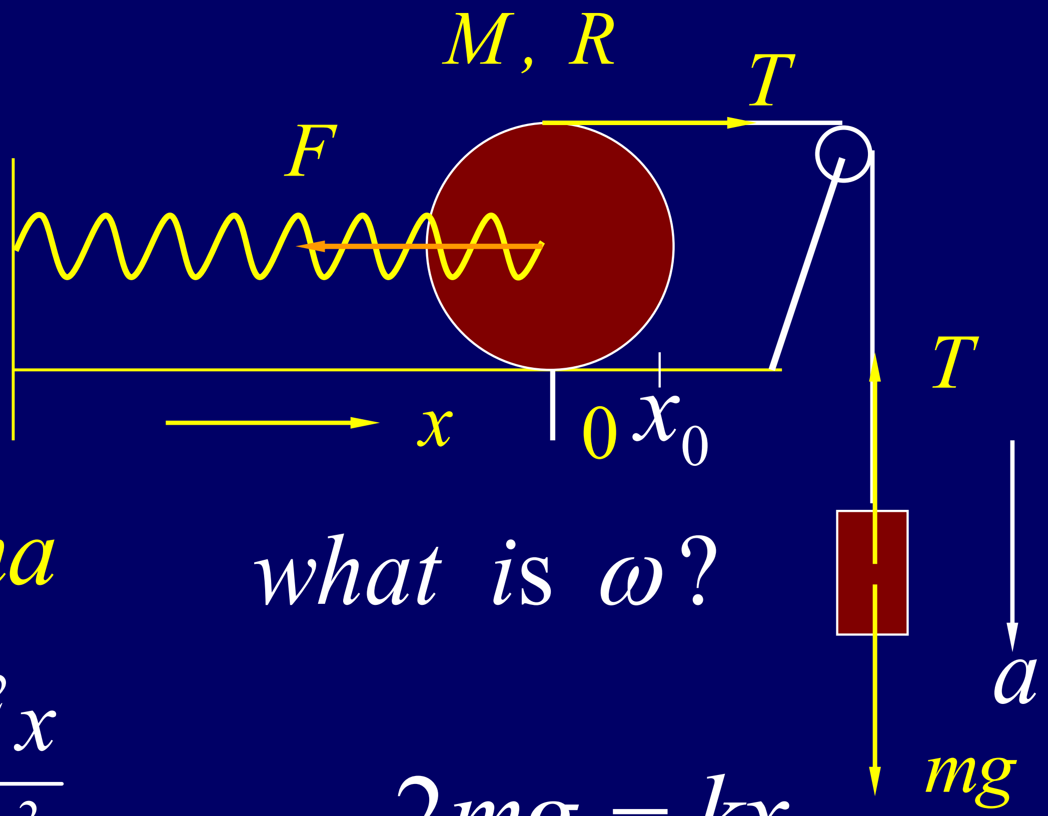
$$\frac{d^2x'}{dt^2} + \omega^2 x' = 0$$

$$what\ \ is\ \ \omega? \qquad \omega^2 = \frac{k}{m}$$

$$x' = x'_m \cos(\omega t + \phi)$$

$$mg - T = ma$$

$$2T - kx = \frac{3}{2} MR\alpha$$



$$2mg - kx = \frac{3}{2} MR\alpha + 2ma$$

$$what \;\; is \;\; \omega ?$$

$$R\alpha = a_{cm} = \frac{d^2x}{dt^2} \qquad a = 2\frac{d^2x}{dt^2}$$

$$2mg = kx_0$$

$$2mg - kx = (4m + \frac{3}{2}M)\frac{d^2x}{dt^2}$$

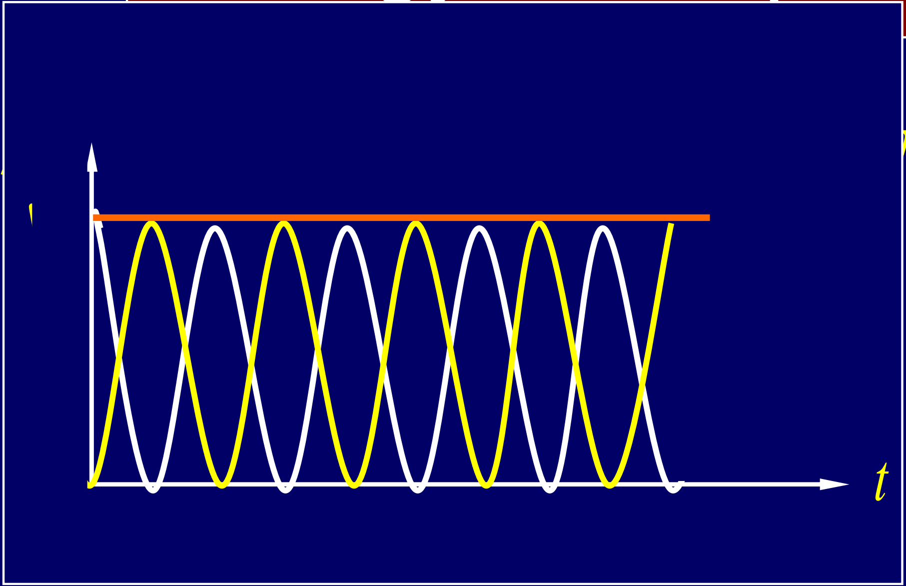
$$-k(x - x_0) = (4m + \frac{3}{2}M)\frac{d^2x}{dt^2}$$

$$\boxed{\omega^2 = \frac{k}{4m + \frac{3}{2}M}}$$

$$\frac{d^2x'}{dt^2} + \omega^2 x' = 0$$

$$x' = x'_m \cos(\omega t + \phi)$$

Energy of Simple Harmonic Oscillations



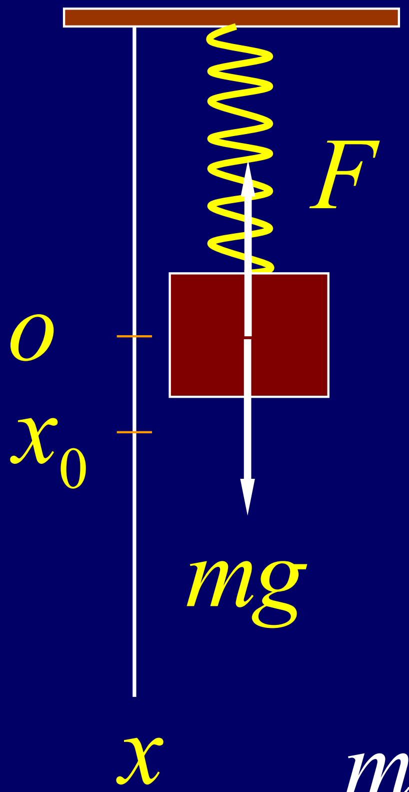
$$U(x) = \frac{1}{2}kx^2 \qquad K = \frac{1}{2}mv^2$$

$$U = \frac{1}{2}kx_m^2 \cos^2(\omega t + \phi)$$

$$K = \frac{1}{2}mx_m^2\omega^2 \sin^2(\omega t + \phi)$$

$$\omega^2 = \frac{k}{m} \qquad K = \frac{1}{2}kx_m^2 \sin^2(\omega t + \phi)$$

$$E = K + U = \frac{1}{2}kx_m^2 = const.$$



$$-mgx+\frac{1}{2}kx^2+\frac{1}{2}m\left(\frac{dx}{dt}\right)^2=E$$

$$\frac{dE}{dt}=0 \qquad -mg\frac{dx}{dt}+kx\frac{dx}{dt}+m\left(\frac{dx}{dt}\right)\left(\frac{d^2x}{dt^2}\right)=0$$

$$-mg+kx+m\frac{d^2x}{dt^2}=0$$

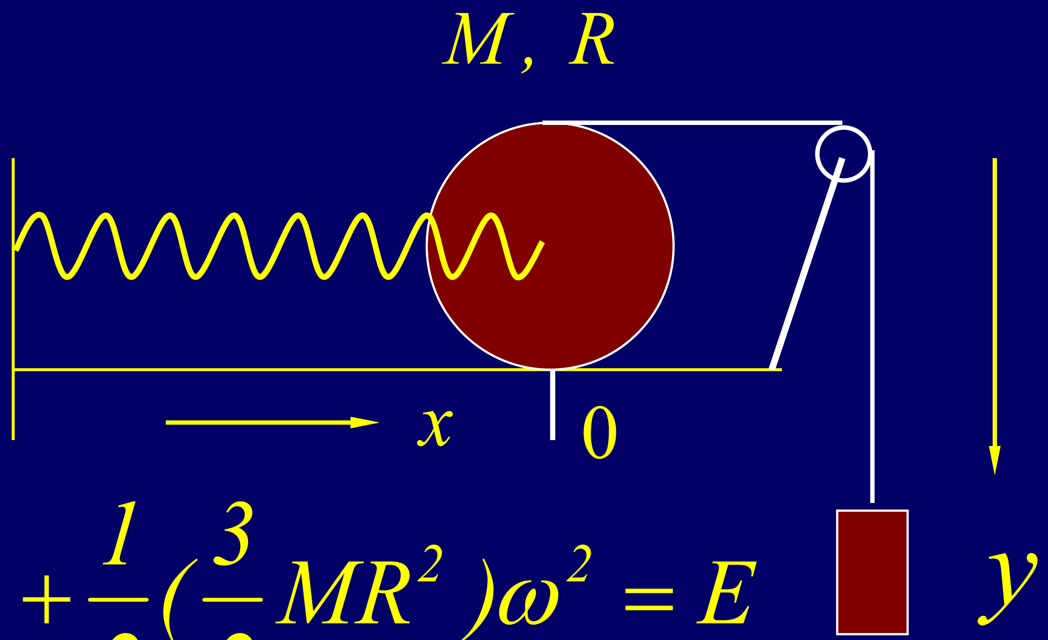
$$mg=kx_0$$

$$let \quad x-x_0=x' \qquad \frac{d^2x'}{dt^2}+\frac{k}{m}x'=0$$

$$let \quad \omega=\sqrt{\frac{k}{m}} \qquad x'=x'_m\cos(\omega t+\phi)$$

$$R\omega = \frac{dx}{dt} \qquad y = 2x$$

$$\frac{dy}{dt} = 2\frac{dx}{dt}$$



$$-mgy + \frac{1}{2}kx^2 + \frac{1}{2}m(\frac{dy}{dt})^2 + \frac{1}{2}(\frac{3}{2}MR^2)\omega^2 = E$$

$$-2mgx + \frac{1}{2}kx^2 + 2m(\frac{dx}{dt})^2 + \frac{1}{2}(\frac{3}{2}M)(\frac{dx}{dt})^2 = E$$

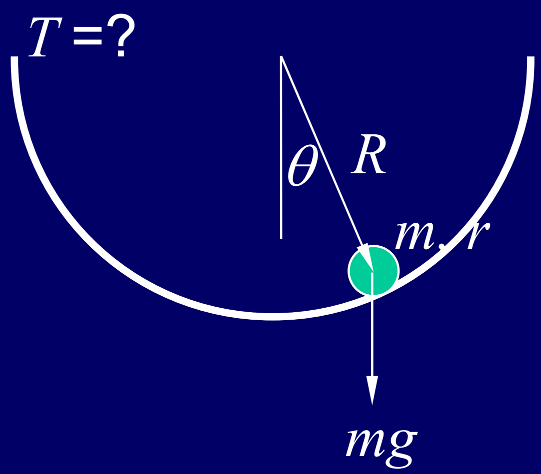
$$2mg = kx_0 \qquad k(x-x_0) + (4m + \frac{3}{2}M)\frac{d^2(x-x_0)}{dt^2} = 0$$

$$x' = x - x_0 \qquad \omega^2 = \frac{k}{4m + \frac{3}{2}M} \qquad \frac{d^2x'}{dt^2} + \omega^2 x' = 0$$

$$x' = x'_m \cos(\omega t + \phi)$$

$r << R$

Rolls without slipping



Force analyse

Energy analyse

$$-mgr\theta = \frac{3}{2}mr^2\frac{R}{r}\frac{d^2\theta}{dt^2}$$

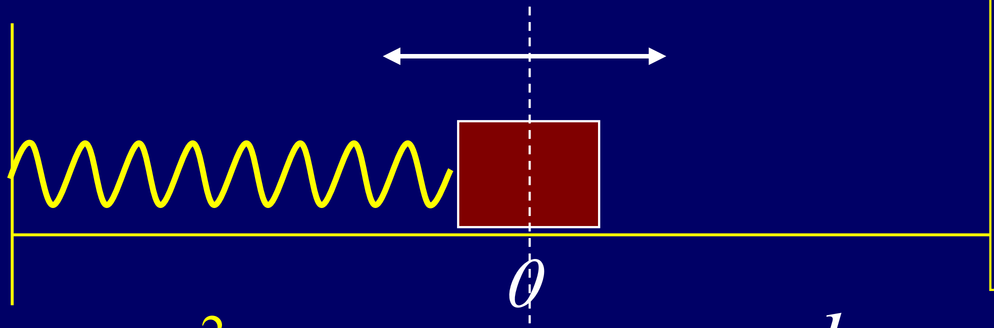
$$\frac{d^2\theta}{dt^2} + \frac{2g}{3R}\theta = 0 \qquad \omega = \sqrt{\frac{2g}{3R}} \qquad T = 2\pi\sqrt{\frac{3R}{2g}}$$

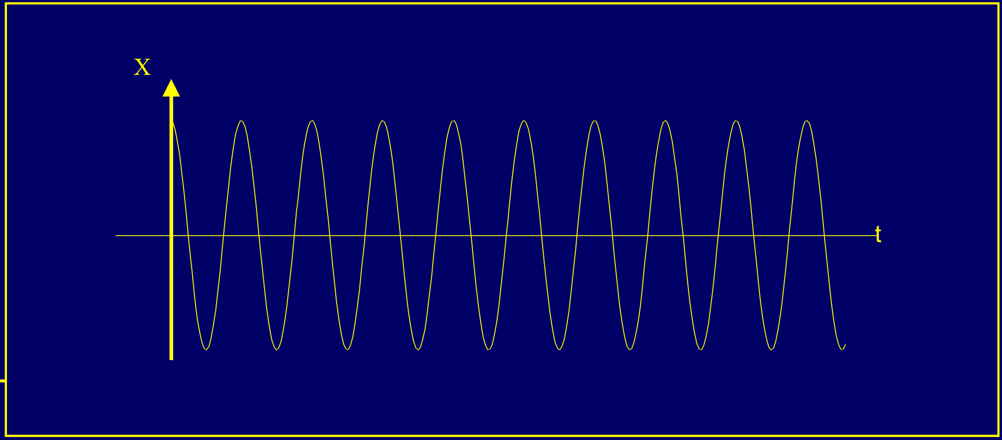
$$\frac{1}{2}(\frac{3}{2}mr^2)(\frac{d\phi}{dt})^2 + mgR(1-\cos\theta) = E$$

$$\frac{3}{2}mr^2\frac{R^2}{r^2}\frac{d\theta}{dt}\frac{d^2\theta}{dt^2} + mgR\theta\frac{d\theta}{dt} = 0$$

$$\frac{d^2\theta}{dt^2} + \frac{2g}{3R}\theta = 0 \qquad \omega = \sqrt{\frac{2g}{3R}} \qquad T = 2\pi\sqrt{\frac{3R}{2g}}$$

Simple Harmonic Oscillation





$$m\frac{d^2x}{dt^2} = -kx \qquad \omega^2 = \frac{k}{m}$$

ω depends on the system

x_m and ϕ depend on the initial condition

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

$$U + K = E = const.$$

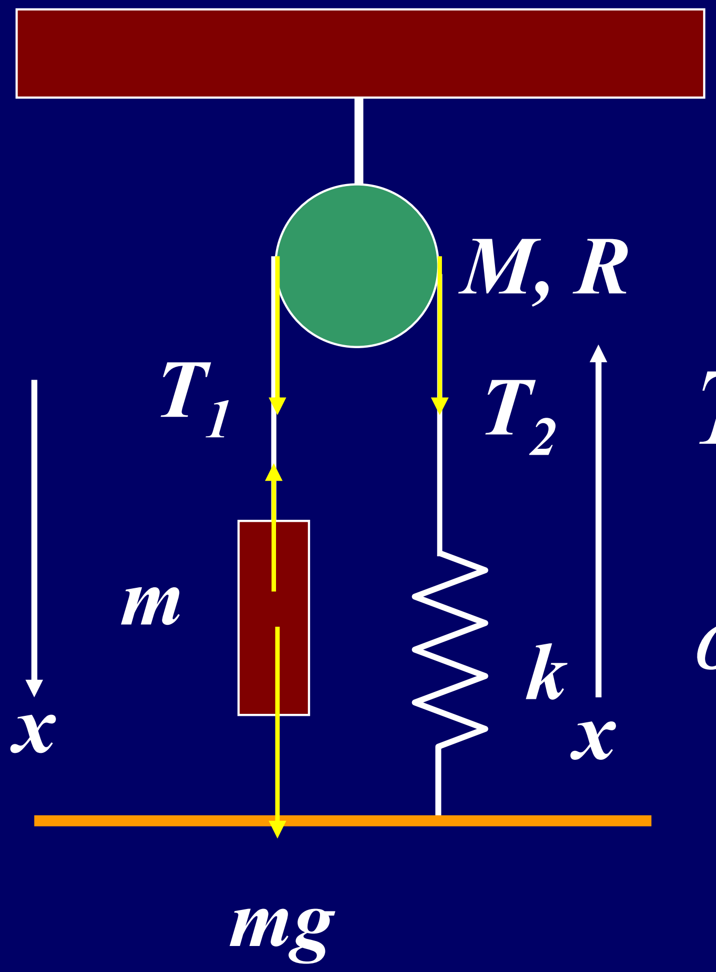
$$\frac{1}{2}m\left(\frac{dx}{dt}\right)^2 + \frac{1}{2}kx^2 = E$$

$$x = x_m \cos(\omega t + \phi)$$

$$m\frac{d^2x}{dt^2} + kx = 0$$

$$v_x = -x_m\omega \sin(\omega t + \phi)$$

$$a_x = -x_m\omega^2 \cos(\omega t + \phi)$$



$$\omega = ?$$

$$mg - T_1 = m\frac{d^2x}{dt^2}$$

$$T_1 - T_2 = \frac{1}{2}MR\alpha$$

$$T_2 = kx$$

$$\alpha R = \frac{d^2x}{dt^2}$$

$$mg - kx = (\frac{1}{2}M + m)\frac{d^2x}{dt^2}$$

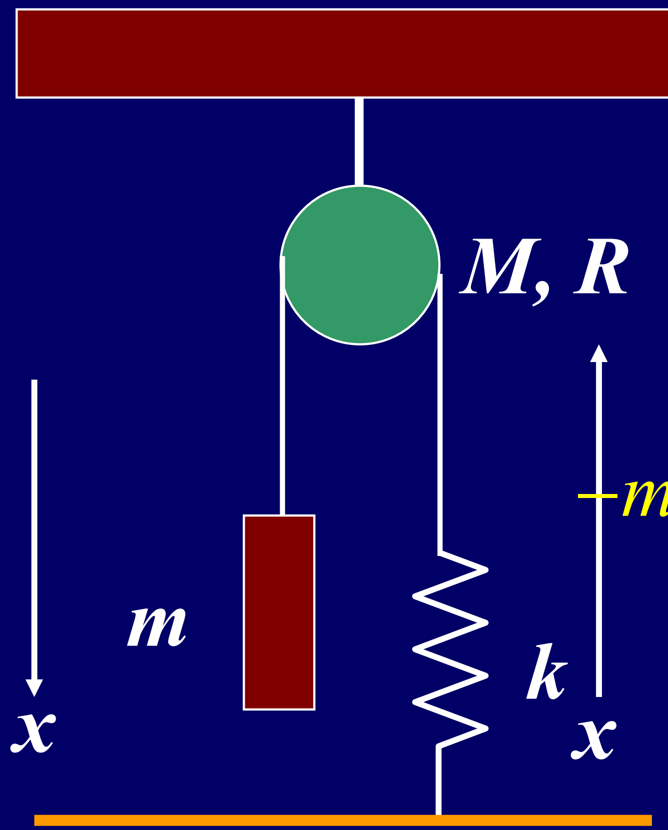
$$mg = kx_0$$

$$-k(x - x_0) = (\frac{1}{2}M + m)\frac{d^2(x - x_0)}{dt^2}$$

$$\omega^2 = \frac{k}{\frac{1}{2}M + m}$$

$$\frac{d^2x'}{dt^2} + \omega^2 x' = 0$$

$$x' = x_m \cos(\omega t + \varphi)$$



$$\omega R = \frac{dx}{dt}$$

$$-mgx + \frac{1}{2}kx^2 + \frac{1}{2}m\left(\frac{dx}{dt}\right)^2 + \frac{1}{2}\left(\frac{1}{2}M\right)\left(\frac{dx}{dt}\right)^2 = E$$

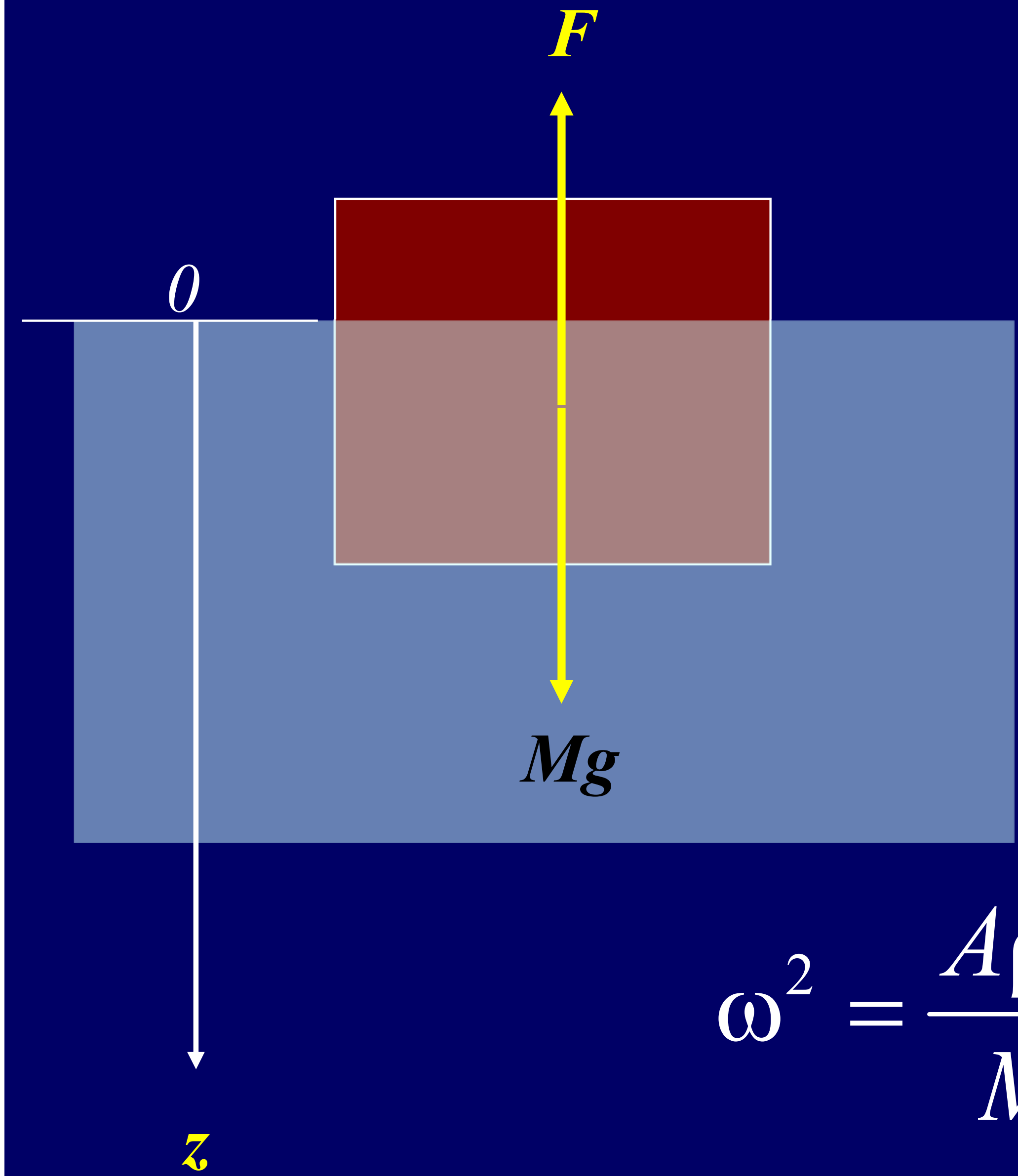
$$-mg\frac{dx}{dt} + kx\frac{dx}{dt} + m\frac{dx}{dt}\left(\frac{d^2x}{dt^2}\right) + \frac{1}{2}M\frac{dx}{dt}\left(\frac{d^2x}{dt^2}\right) = 0$$

$$-mg + kx + m\left(\frac{d^2x}{dt^2}\right) + \frac{1}{2}M\left(\frac{d^2x}{dt^2}\right) = 0$$

$$mg = kx_0 \qquad -k(x - x_0) = \left(\frac{1}{2}M + m\right)\frac{d^2(x - x_0)}{dt^2}$$

$$\omega^2 = \frac{k}{\frac{1}{2}M + m} \qquad \frac{d^2x'}{dt^2} + \omega^2 x' = 0$$

$$x' = x_m \cos(\omega t + \varphi)$$



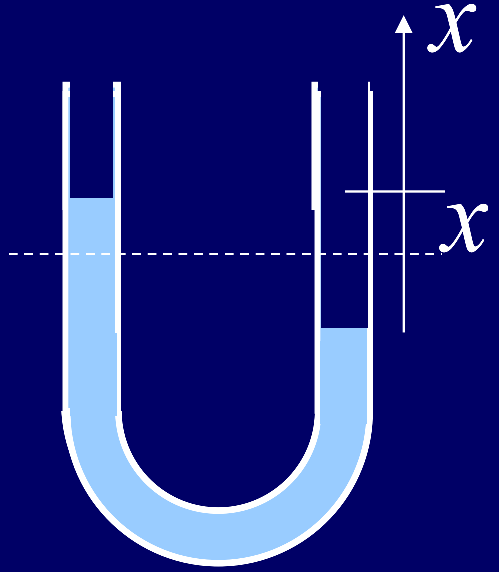
$$\Delta F = -zA\rho g$$

$$M\frac{d^2 z}{dt^2} = -zA\rho g$$

$$\frac{d^2 z}{dt^2} + \frac{A\rho g}{M} z = 0$$

$$\omega^2 = \frac{A\rho g}{M} \qquad \frac{d^2 z}{dt^2} + \omega^2 z = 0$$

$$z = z_m \cos(\omega t + \phi)$$



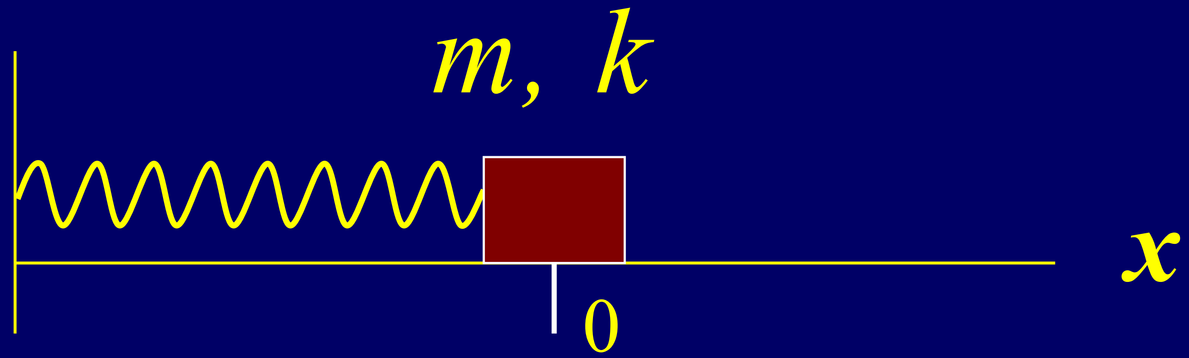
$$\Delta p = 2g\rho x$$

$$F = -2Ag\rho x = m\frac{d^2x}{dt^2}$$

$$\frac{d^2x}{dt^2} + \frac{2A\rho g}{m}x = 0$$

$$\frac{2A\rho g}{m} = \omega^2 \qquad \frac{d^2x}{dt^2} + \omega^2 x = 0$$

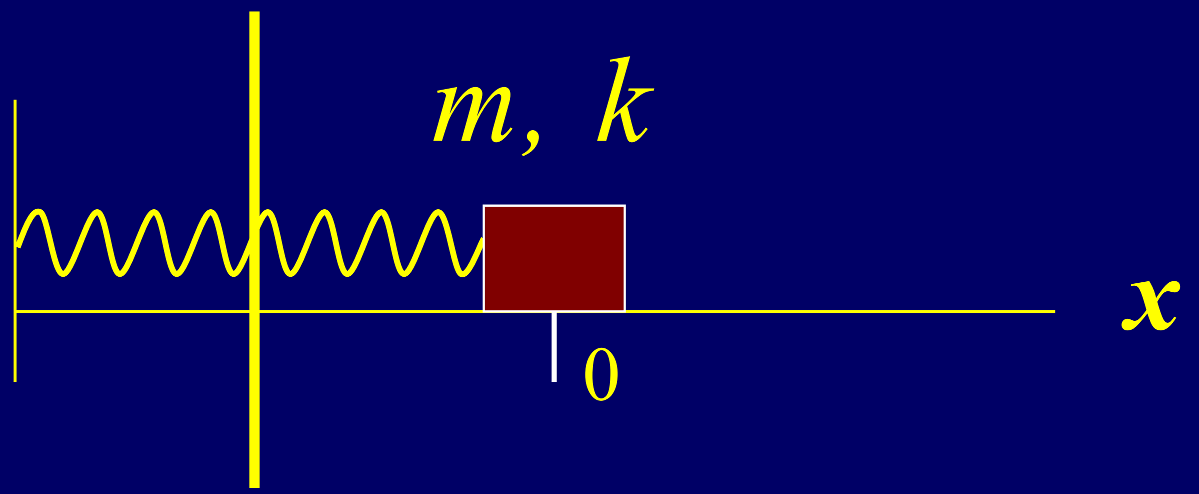
$$x = x_m \cos(\omega t + \phi)$$



$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

$$\omega = \sqrt{\frac{k}{m}}$$

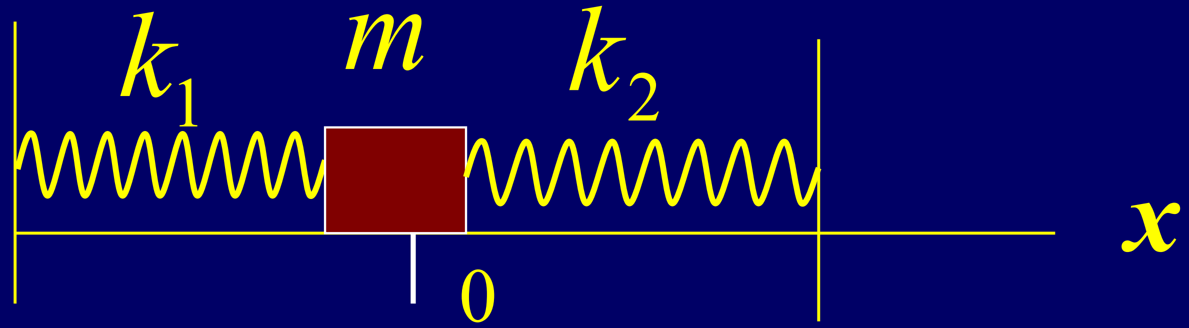
$$x = x_m\, cos(\omega t + \phi)$$



$$F = -kx \qquad F = -E\frac{\Delta L}{L} \qquad k = \frac{E}{L}$$

$$L' = \frac{L}{2} \qquad k' = 2k \qquad \omega' = \sqrt{\frac{k'}{m}}$$

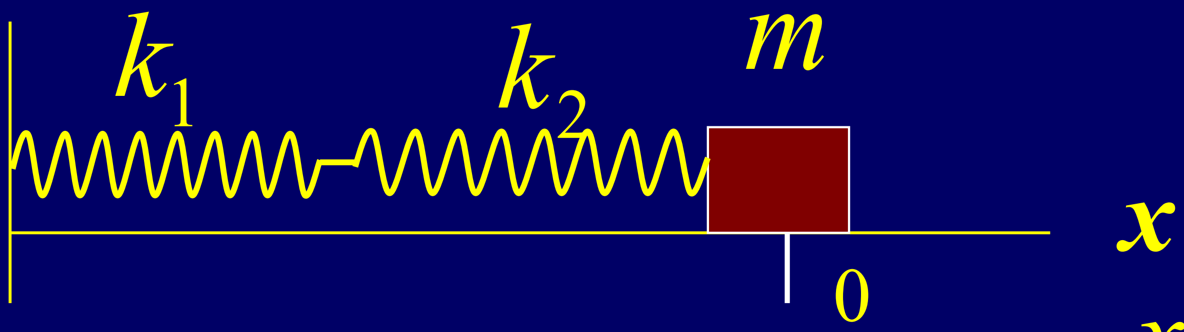
$$\omega' = \sqrt{2}\,\omega$$



$$F=-k_1x-k_2x=-(k_1+k_2)x=-k'x$$

$$k'=k_1+k_2$$

$$\omega'=\sqrt{\frac{k_1+k_2}{m}}=\sqrt{\frac{k_1}{m}+\frac{k_2}{m}}=\sqrt{\omega_1^2+\omega_2^2}$$



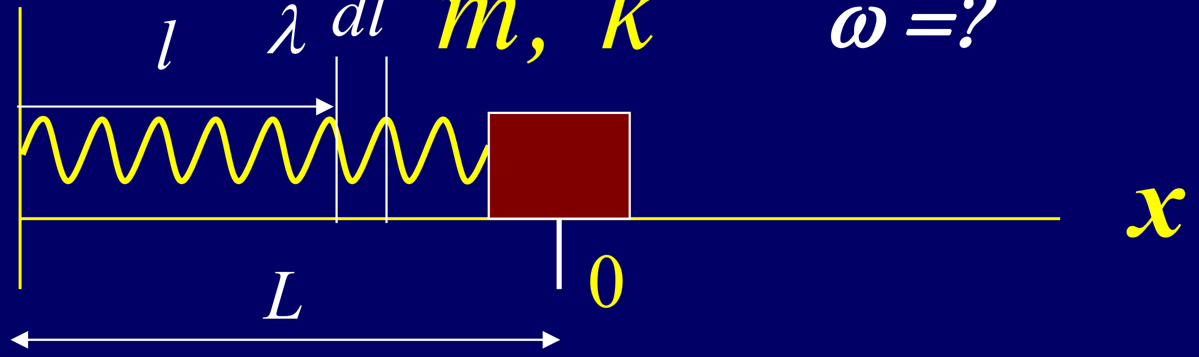
$$F = -k_1 x_1 = -k_2 x_2 \qquad \frac{x_1}{x_2} = \frac{k_2}{k_1}$$

$$x = x_1 + x_2 = (1 + \frac{k_2}{k_1}) x_2 \qquad \frac{d^2 x}{dt^2} = (1 + \frac{k_2}{k_1}) \frac{d^2 x_2}{dt^2}$$

$$\frac{d^2 x_2}{dt^2} + \omega^2 x_2 = 0 \qquad \omega^2 = \frac{k_1 k_2}{m(k_1 + k_2)}$$

$$x_2 = x_m \cos(\omega t + \varphi) \qquad \frac{1}{\omega^2} = \frac{1}{\omega_1^2} + \frac{1}{\omega_2^2}$$

$m,\ k$ $\qquad \omega = ?$



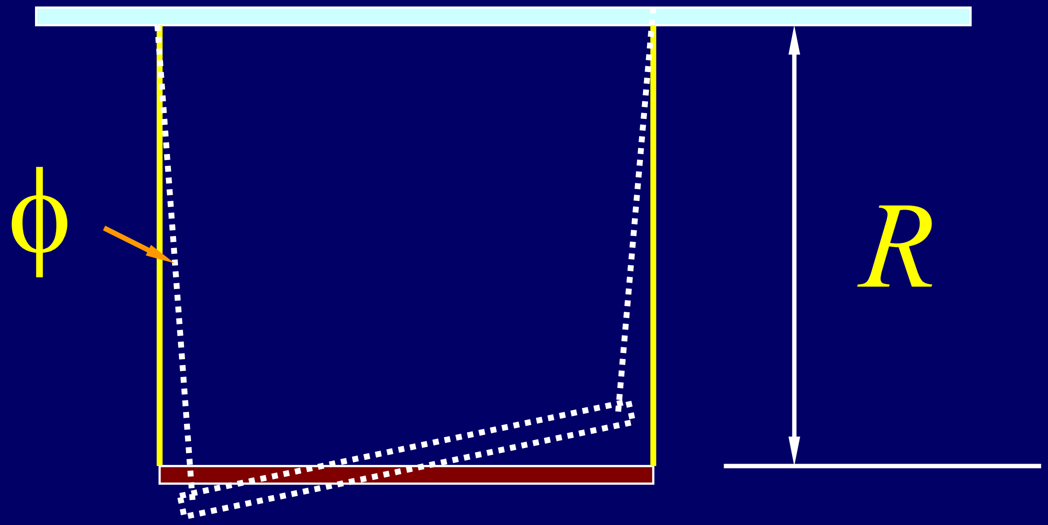
$$E = \frac{1}{2}kx^2 + \frac{1}{2}(m + \frac{1}{3}\lambda L)(\frac{dx}{dt})^2$$

$$\omega^2 = \frac{k}{m + \frac{1}{3}\lambda L}$$

$$kx + (m + \frac{1}{3}\lambda L)\frac{d^2x}{dt^2} = 0$$

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

$$x = x_m \cos(\omega t + \varphi)$$



$$m,\quad 2l$$

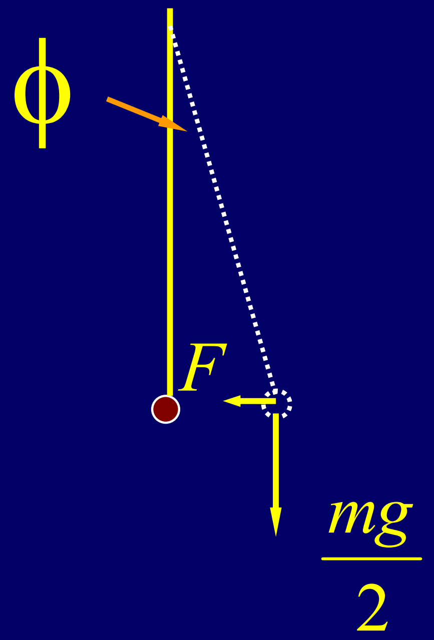
$$T = ?$$

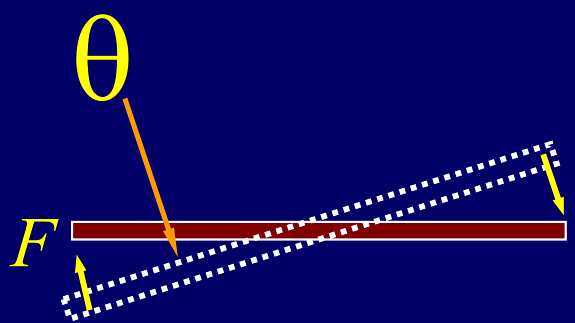
$$F = \frac{mg}{2} tg\phi$$

$$\tau = -2Fl = -mgl\phi$$

$$-mgl\phi = \frac{1}{3} ml^2 \frac{d^2\theta}{dt^2}$$

$$R\varphi = l\theta \qquad -g\theta = \frac{1}{3} R \frac{d^2\theta}{dt^2}$$

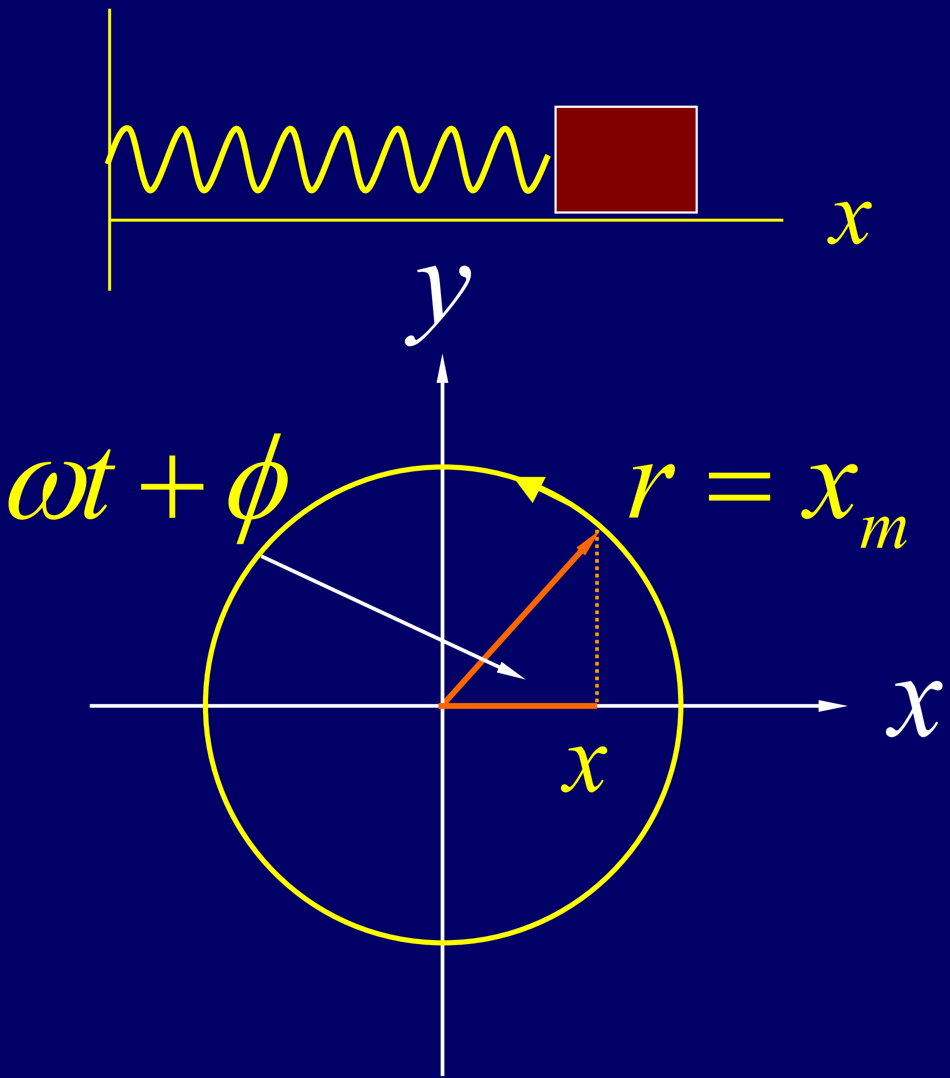




$$\frac{d^2\theta}{dt^2} + \frac{3g}{R}\theta = 0$$

$$\omega = \sqrt{\frac{3g}{R}} \qquad T = 2\pi\sqrt{\frac{R}{3g}}$$

Simple Harmonic Oscillations and Uniform Circular Motion

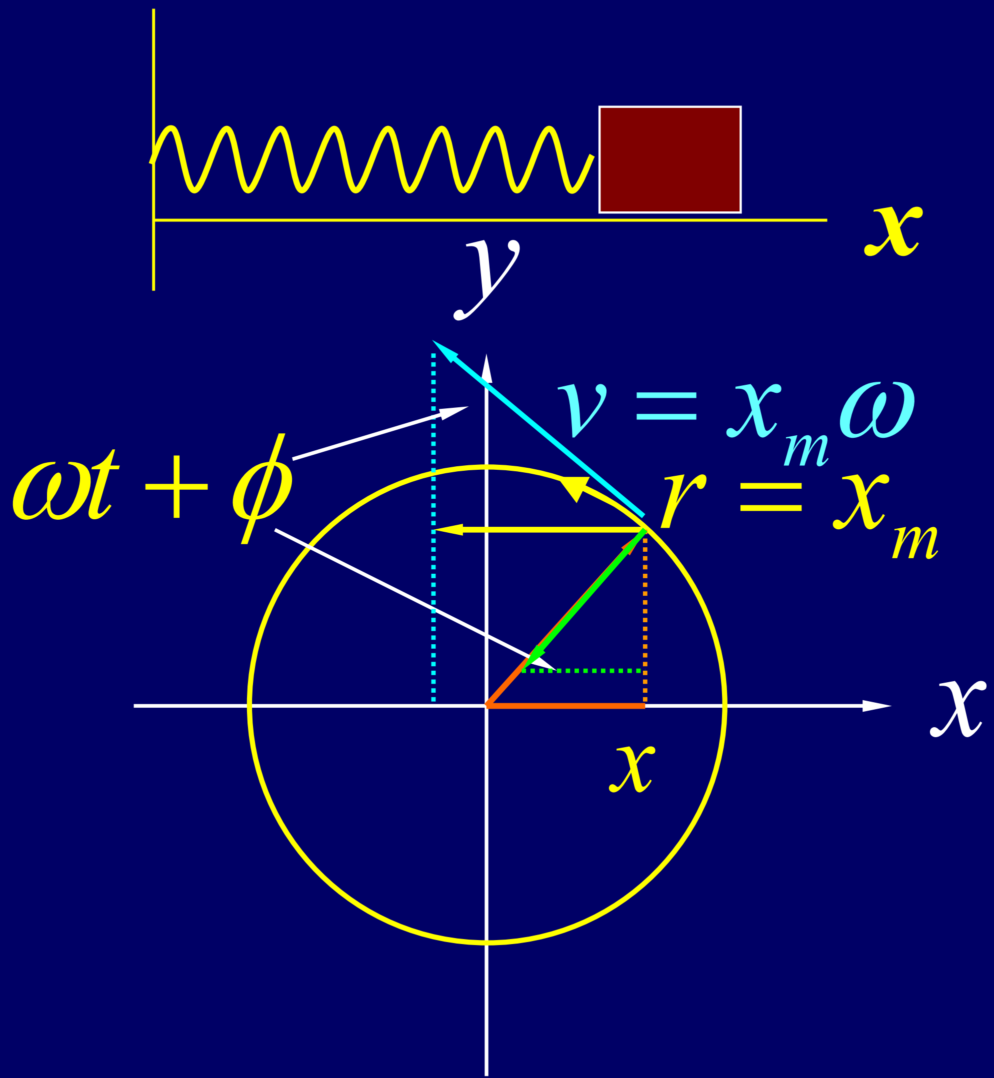


$$x = x_m \cos(\omega t + \phi)$$

$$v_x = -x_m \omega \sin(\omega t + \phi)$$

$$a_x = -x_m \omega^2 \cos(\omega t + \phi)$$

Simple Harmonic Oscillations and Uniform Circular Motion



$$x = x_m \cos(\omega t + \phi)$$

$$v_x = -x_m \omega \sin(\omega t + \phi)$$

$$a_x = -x_m \omega^2 \cos(\omega t + \phi)$$

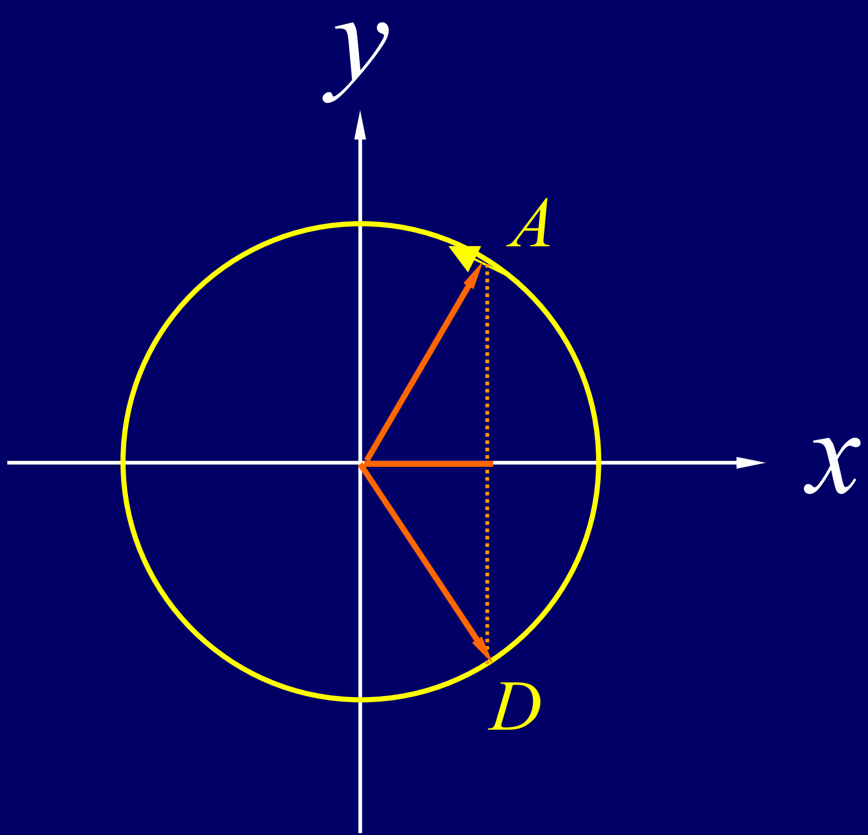
$$a = \frac{(x_m \omega)^2}{x_m} = x_m \omega^2$$

ω angular frequency

angular velocity

A particle with velocity v_0 $(v_0 > 0)$, *at the half point to the maxmum point. After 2 second it comes back this point. What is* x_m, ω *and* ϕ?

Begin at *D* point $\phi = -\frac{\pi}{3}$ $\omega \cdot 2 = \frac{2\pi}{3}$ $\omega = \frac{\pi}{3}$



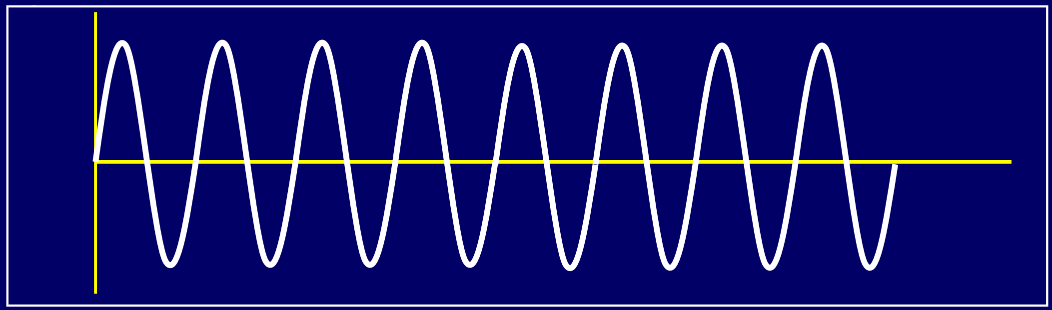
$$v_x = -x_m \omega \sin(\omega t + \phi)$$

$$v_0 = -x_m \frac{\pi}{3} \sin\left(-\frac{\pi}{3}\right)$$

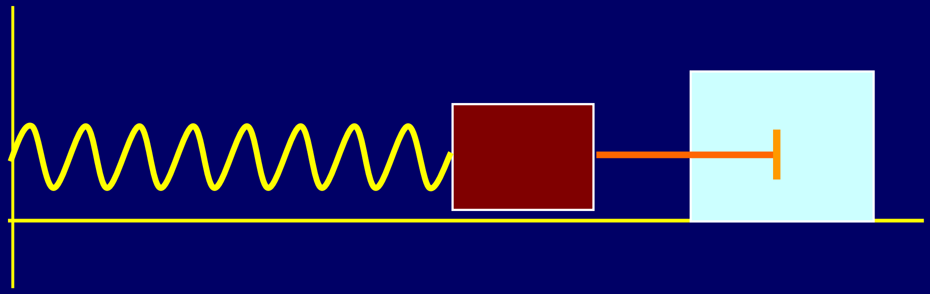
$$x_m = v_0 \frac{2\sqrt{3}}{\pi}$$

Damped Harmonic Motion

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$



$$x = x_m \cos(\omega t + \phi)$$



$$m\frac{d^2x}{dt^2} = -kx - b\frac{dx}{dt}$$

Damped Harmonic Motion.

- The general solution of this equation of motion is

$$x(t) = Ae^{i\omega t}$$

- If we substitute this solution in the equation of motion we find

$$-\omega^2 Ae^{i\omega t} + i\omega\frac{b}{m}Ae^{i\omega t} + \frac{k}{m}Ae^{i\omega t} = 0$$

- In order to satisfy the equation of motion, the angular frequency must satisfy the following condition:

$$\omega^2 - i\omega\frac{b}{m} - \frac{k}{m} = 0$$

Damped Harmonic Motion.

- We can solve this equation and determine the two possible values of the angular velocity:

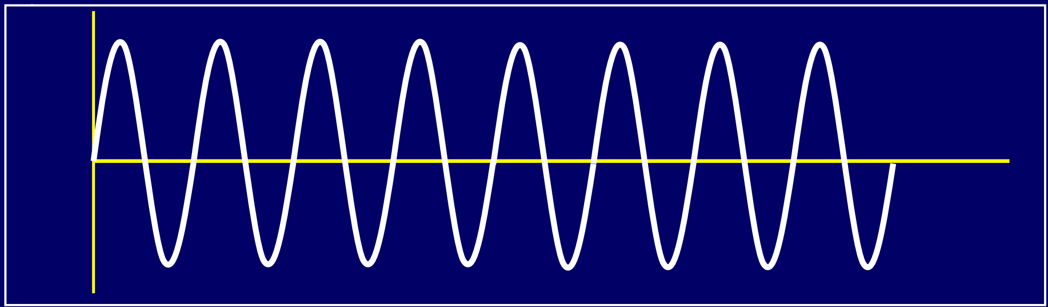
$$\omega = \left(i\frac{b}{2m} \pm \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}} \right)$$

- The solution to the equation of motion is thus given by

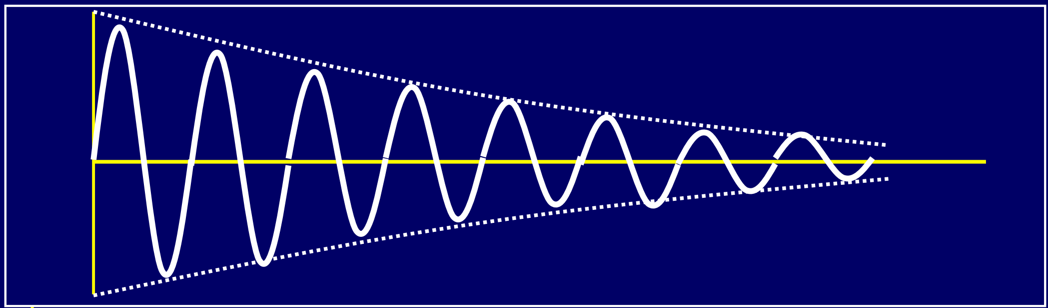
$$x(t) \sim x_m \underbrace{e^{-\frac{bt}{2m}}}_{\text{Damping Term}} \underbrace{\cos(\omega' t)}_{\text{SHM Term}} \qquad \omega' = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}$$

Damped Harmonic Motion

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$



$$x = x_m \cos(\omega t + \phi)$$



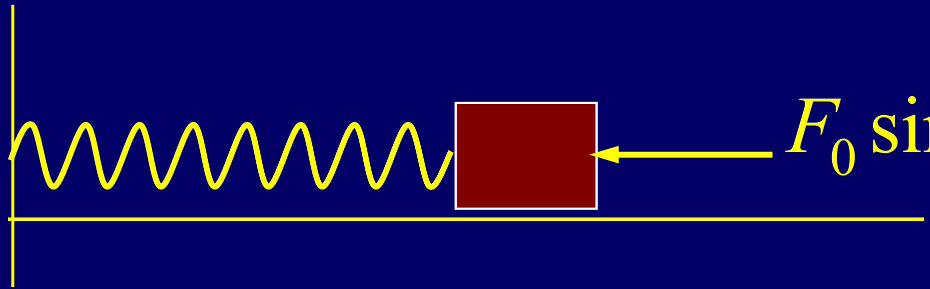
$$m\frac{d^2x}{dt^2} = -kx - b\frac{dx}{dt}$$

$$x = x_m e^{-bt/2m} \cos(\omega' t + \phi)$$

$$\omega' = \sqrt{\frac{k}{m} - \left(\frac{b}{2m}\right)^2}$$

Forced Oscillations

Resonance



$$\frac{d^2x}{dt^2} + \omega_0{}^2 x = F_0 \sin(\omega t)$$

Driven Harmonic Motion.

- Consider the general solution

$$x(t) = A\cos(\omega t + \phi)$$

- The parameters in this solution must be chosen such that the equation of motion is satisfied. This requires that

$$-\omega^2 A\cos(\omega t + \phi) + \omega_0{}^2 A\cos(\omega t + \phi) - F_0 \sin(\omega t) = 0$$

- This equation can be rewritten as

$$\left(\omega_0{}^2 - \omega^2\right) A\cos(\omega t)\cos(\phi) -$$
$$\left(\omega_0{}^2 - \omega^2\right) A\sin(\omega t)\sin(\phi) - F_0 \sin(\omega t) = 0$$

Driven Harmonic Motion.

- Our general solution must thus satisfy the following condition:

$$\left(\omega_0{}^2-\omega^2\right)A\cos\left(\omega t\right)\cos\left(\phi\right)-\left\{\left(\omega_0{}^2-\omega^2\right)A\sin\left(\phi\right)-F_0\right\}\sin\left(\omega t\right)=0$$

- Since this equation must be satisfied at all time, we must require that the coefficients of cos(ωt) and sin(ωt) are 0. This requires that

$$\left(\omega_0{}^2-\omega^2\right)A\cos\left(\phi\right)=0$$

and

$$\left(\omega_0{}^2-\omega^2\right)A\sin\left(\phi\right)-F_0=0$$

Driven Harmonic Motion.

- The interesting solutions are solutions where $A \neq 0$ and $\omega \neq \omega_0$. In this case, our general solution can only satisfy the equation of motion if

$$\cos(\phi) = 0$$

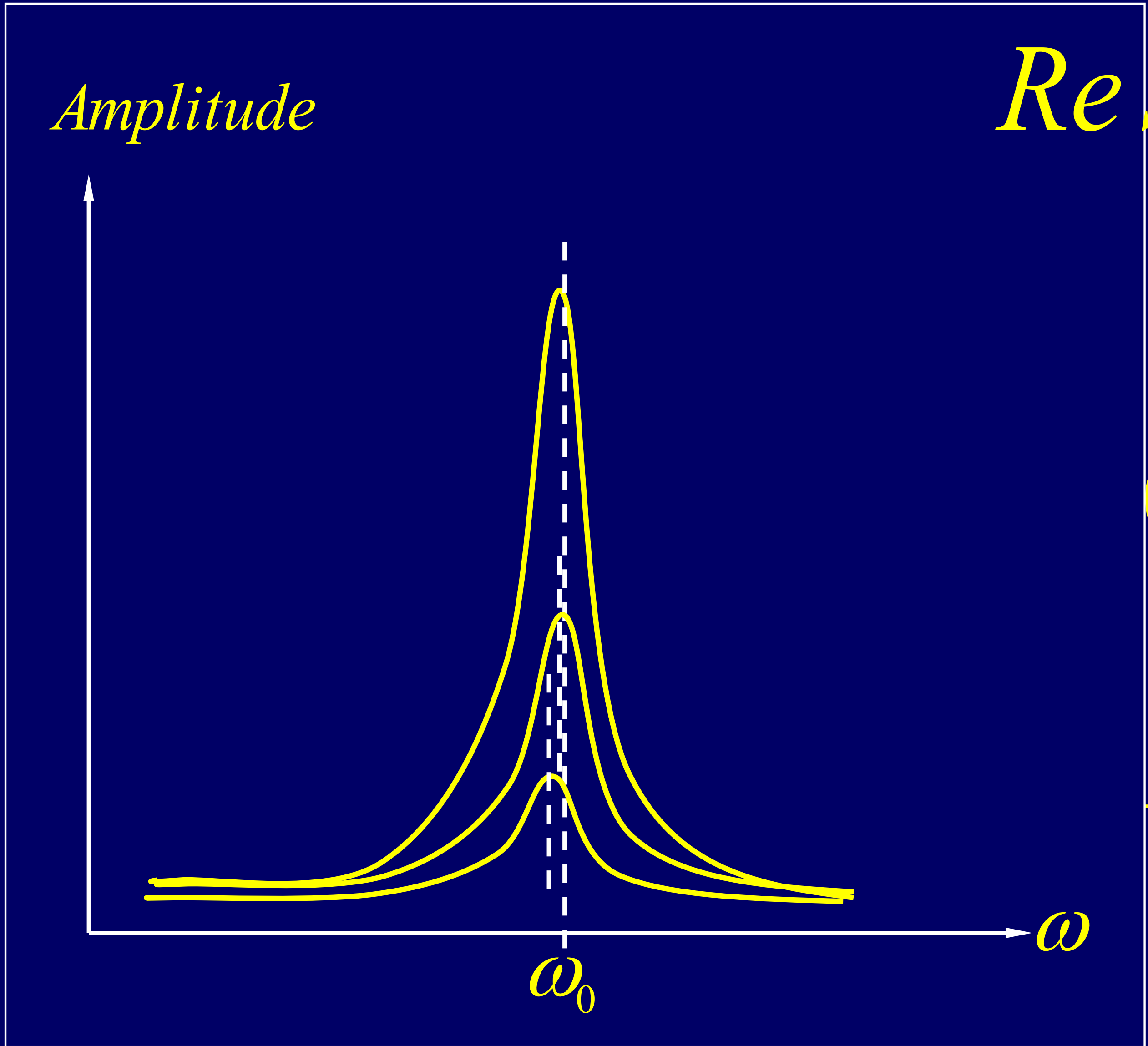
and

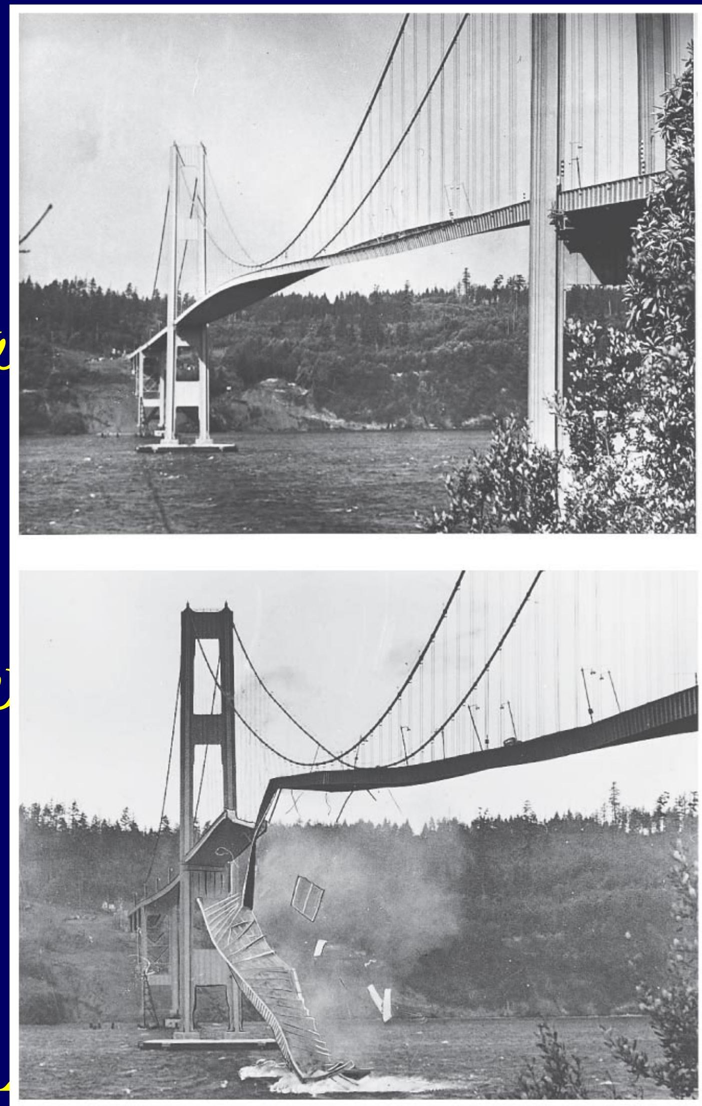
$$\left(\omega_0{}^2 - \omega^2\right) A \sin(\phi) - F_0 = \left(\omega_0{}^2 - \omega^2\right) A - F_0 = 0$$

- The amplitude of the motion is thus equal to

$$A = \frac{F_0}{\left(\omega_0{}^2 - \omega^2\right)}$$

Resonance

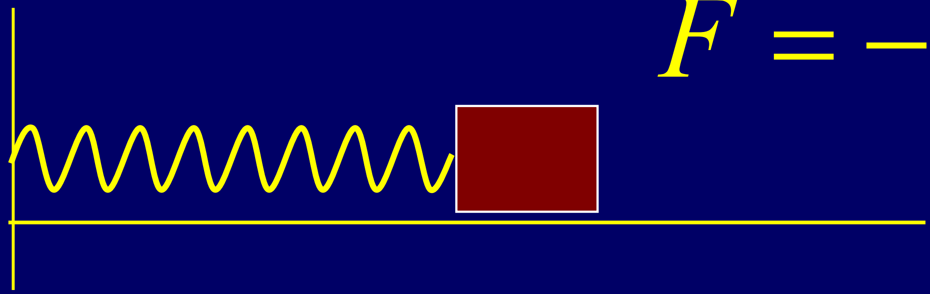




$$A = \frac{F_0}{\left(\omega_0^{\,2} - \omega^2\right)}$$

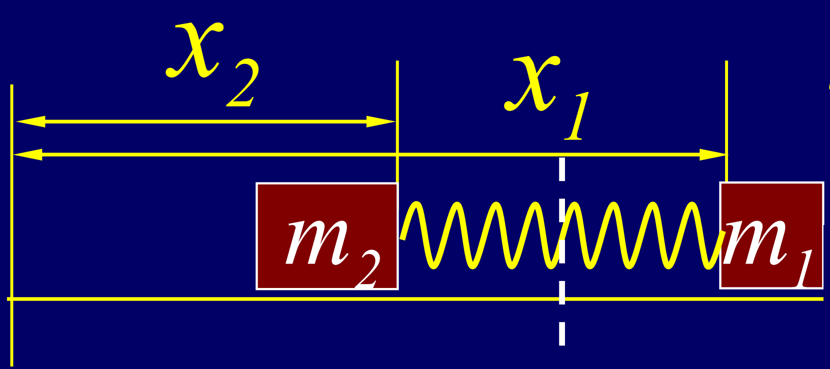
$$x(t) =$$

Two-Body Oscillations



$$F = -kx \qquad F = -k(x_1 - x_2 - L)$$

$$m_1 m_2 \frac{d^2 x_1}{dt^2} = -m_2 k x$$



$$if \quad m_2 >> m_1 \qquad \frac{m_1 m_2}{(m_1 + m_2)} = m^* = m_1$$

$$if \quad m_2 = m_1 \qquad \frac{m_1 m_2}{(m_1 + m_2)} = m^* = \frac{m_1}{2}$$

$$\frac{d^2}{dt^2} x + \omega^2 x = 0 \qquad \omega^2 = \frac{2k}{m_1} \qquad \frac{d^2}{dt^2} x + \frac{k}{m^*} x = 0$$

CHAP. 17

Exercises

P395:25, 31, 36, 37, 38, 42, 43, 44

Problems

P397: 11, 16